

From mPL to Shapley Attribution: Quantifying Per-Cue Location-Privacy Leakage in Images

Technical note — GeoBayes project

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Abstract. A photograph leaks its capture location through many visual cues. We quantify *how much each cue leaks* with respect to a concrete adversary. Leakage of a cue subset is measured by mPL, a distance-normalized posterior-leakage metric. Because masking effects are non-additive (cues overlap or back each other up), single-cue mPL is a biased per-cue estimate; we treat leakage as a set function and attribute it with the Shapley value, whose efficiency axiom makes per-cue shares sum exactly to the total removable leakage of the identified cue set. The pipeline (GPT-4o cue naming, SAM 3 masks, GeoRanker adversary) is evaluated on 100 im2gps3k images.

1 Problem Statement

Let I be an image with m identified cues $N = \{1, \dots, m\}$, each with a pixel mask, and an adversary who forms a posterior over a fixed candidate set. Two questions: how much belief movement is caused by masking any cue *subset* (measurement, §2), and how to divide the total leakage among individual cues when effects do not add up (attribution, §4).

2 Privacy Criterion and Adversary Model

2.1 Setup

Let $X \in \mathcal{X}$ denote the capture location of image I , where $(\mathcal{X}, d_X)$ is a finite metric space of candidate locations and d_X is the haversine distance. We instantiate $\mathcal{X}$ with the $n = 138$ city-level ground-truth labels of our im2gps3k subset (administrative aliases within 2 km merged — only City of Westminster / Greater London, 1.3 km — leaving 137 points; genuinely distinct nearby places are kept separate). For a cue subset $S \subseteq N$, the masked image $I \ominus S$ — gray-filling the union of the SAM 3 masks of the cues in S — plays the role of the sanitized release.

Adversary model. The criterion below is adversary-agnostic; for empirical auditing we instantiate the adversary as a neural posterior estimator with scoring function $s_\theta(x; I)$ — the GeoRanker reward model (Qwen2-VL-7B + LoRA + value head), whose score increases with geographic closeness between the depicted scene and candidate x . Combining the scores by a softmax yields a *normalized adversarial belief* (we avoid calling it a calibrated Bayesian posterior; see §2.3): ($\tau = 1$; deterministic at fixed input):

$$\Pr_\theta(X = x_i | I) = \frac{\exp(s_\theta(x_i; I)/\tau)}{\sum_{x' \in \mathcal{X}} \exp(s_\theta(x'; I)/\tau)}, \quad x_i \in \mathcal{X} \quad (1)$$

2.2 Metric-normalized posterior leakage

Definition 2.1 (mPL, following [4]). For $x_i, x_j \in X$ with $x_i \neq x_j$ and $d_X(x_i, x_j) > 0$, the pairwise leakage of cue subset S is

$$\text{mPL}(x_i, x_j; S) = \frac{1}{d_X(x_i, x_j)} \left| \ln \frac{\Pr_\theta(X=x_i | I)}{\Pr_\theta(X=x_j | I)} - \ln \frac{\Pr_\theta(X=x_i | I \ominus S)}{\Pr_\theta(X=x_j | I \ominus S)} \right| \quad (2)$$

and the image-level leakage aggregates over the admissible pair set P (pairs with positive posterior mass under both releases):

$$\text{mPL}(S) = \frac{1000}{|P|} \sum_{(i,j) \in P} \text{mPL}(x_i, x_j; S) \quad (3)$$

Here $\Pr_\theta(\cdot | I)$ is the *full-view belief* and $\Pr_\theta(\cdot | I \ominus S)$ the *counterfactual baseline belief* (masked view): mPL measures the distance-normalized odds shift between them; the factor 1000 sets the units to nats per 1000 km. The absolute value makes mPL direction-agnostic. Note $v(N)$, masking all identified cues, is not the total leakage of the image — residual background (climate, terrain, road layout) can still localize; $v(N)$ is the *total removable leakage of the identified cue set*.

2.3 Calibration and temperature invariance

The GeoRanker score is a ranking/distance reward, not a probability-trained log-likelihood, so the softmax is a belief, not a calibrated posterior. This matters because mPL scales inversely with τ : since s cancels additively in the log-odds,

$$p_i(\tau) = \frac{p_i(1)^{1/\tau}}{\sum_j p_j(1)^{1/\tau}}, \quad \text{mPL}(S; \tau) = \frac{1}{\tau} \text{mPL}(S; 1) \quad (4)$$

verified exactly on our data (e.g. $\text{mPL} \times 2.00 / \times 0.50 / \times 0.20$ at $\tau = 0.5 / 2 / 5$). Two consequences. (i) τ is a pure global scale, so *within-image* subset rankings and interaction signs — on which all our conclusions rest — are exactly τ -invariant; absolute magnitudes are reported in $\tau = 1$ units. (ii) For the absolute scale, temperature scaling can be fit on held-out true-label NLL: on a 47/48 split the optimum is $\tau^* = 0.95$ (near 1), with test NLL 2.21→2.19 and Brier 0.643→0.639, confirming $\tau = 1$ is already near-optimal. We therefore report a *score-induced belief-shift attribution* rather than a strictly Bayesian posterior-leakage guarantee.

3 Non-Additivity: Leakage as a Set Function

Define the leakage set function on cue subsets:

$$v(S) = \text{mPL}(S), \quad v(\emptyset) = 0, \quad S \subseteq N = \{1, \dots, m\} \quad (5)$$

Empirically $v(S \cup T) \neq v(S) + v(T)$, in both directions. **Backup (super-additive):** redundant cues compensate for each other, so $v(\{k\}) \approx 0$ for every k while $v(N)$ is large (an Okazaki festival scene shows exactly this). **Overlap (sub-additive):** cues carrying overlapping evidence each cause most of the movement alone, so $v(\{1,2\}) < v(\{1\}) + v(\{2\})$ (limestone cliffs vs. rocky shoreline in a Capri photo). Consequently no single masking condition yields a fair per-cue number: the *leave-one-out* marginal $v(N) - v(N \setminus \{k\})$ *under-credits* backup cues, while the *leave-one-in* value $v(\{k\})$ *over-credits* overlapping ones; in our experiments the two disagreed by up to an order of magnitude.

4 Shapley Attribution

Definition 4.1 (Shapley-mPL). The attributed leakage of cue k is its Shapley value under the set function v :

$$\varphi_k = \sum_{S \subseteq N \setminus \{k\}} \frac{|S|! (m - |S| - 1)!}{m!} [v(S \cup \{k\}) - v(S)] \quad (6)$$

φ_k is the average marginal damage of removing cue k over all $m!$ removal orders, and the unique attribution satisfying efficiency, symmetry, null player, and linearity. Efficiency is the property we rely on:

$$\sum_{k=1}^m \varphi_k = v(N) \quad (7)$$

so “cue k contributes $\varphi_k / v(N)$ of the leakage” is well defined. For two-cue images (the median case) the formula collapses to the average of the leave-one-in and leave-one-out readings:

$$\varphi_1 = \frac{1}{2} v(\{1\}) + \frac{1}{2} [v(\{1, 2\}) - v(\{2\})] \quad (8)$$

Pairwise structure is captured by the Shapley interaction index (SII), which averages the second-order difference over all coalition contexts:

$$I_{kl}^{SII} = \sum_{S \subseteq N \setminus \{k, l\}} \frac{|S|! (m - |S| - 2)!}{(m - 1)!} [v(S \cup \{k, l\}) - v(S \cup \{k\}) - v(S \cup \{l\}) + v(S)] \quad (9)$$

with $I^{SII} < 0$ = overlap and > 0 = backup. **Worked example (Capri).** $v(\{\text{cliffs}\}) = 0.088$, $v(\{\text{shoreline}\}) = 0.104$, $v(N) = 0.113$ give $\varphi_{\text{cliffs}} = 0.049$, $\varphi_{\text{shoreline}} = 0.065$ (sum = $v(N)$), and $I = -0.079$ flags the two coastal cues as overlapping. φ_k can legitimately be negative (masking k pushes belief back); such values are reported as measured.

5 Experimental Pipeline

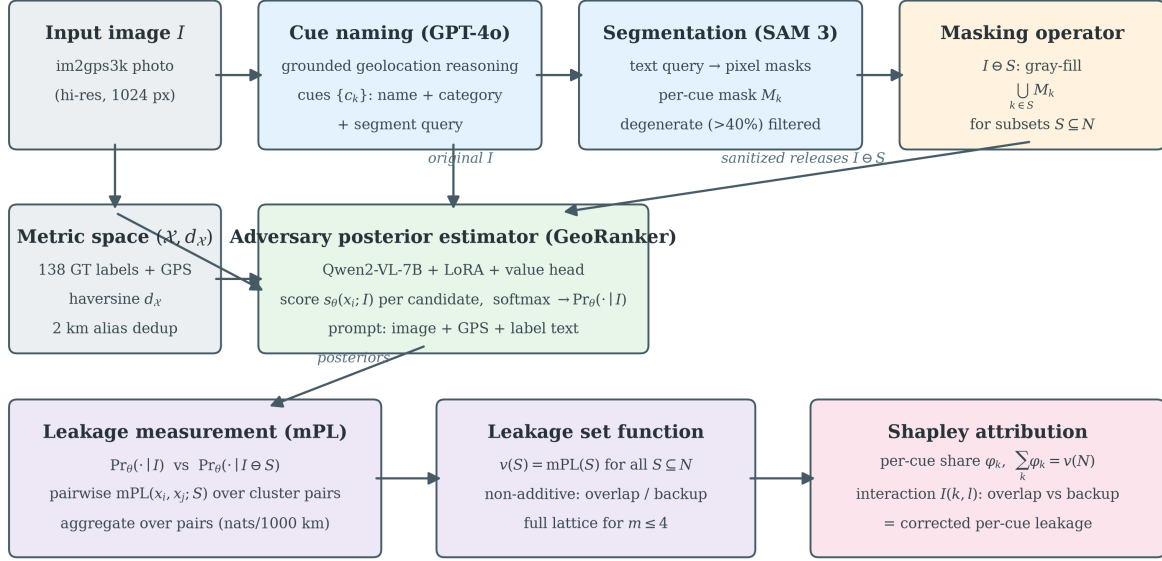


Figure 1: Pipeline. Cues are named by GPT-4o, localized by SAM 3, masked in subsets, and scored by the GeoRanker posterior estimator over the metric space; mPL over all subsets defines v , which the Shapley value attributes to individual cues.

Component	Choice
Dataset	im2gps3k subset, 100 hi-res images
Cue naming	GPT-4o grounded reasoning (name, category, segment query)
Segmentation	SAM 3 text-to-mask; degenerate masks (>40% of frame) excluded
Metric space	138 GT labels + GPS, haversine; 2 km alias dedup (137 points)
Adversary	GeoRanker: Qwen2-VL-7B + LoRA + value head, nf4, deterministic
Elicitation	prompt = image + candidate GPS + label text; softmax $\tau = 1$
Metric	mPL (nats/1000 km), within-image; full posteriors persisted
Attribution	Shapley ϕ + interaction I (full 2^m lattice, $m \leq 5$)

Table 1: Implementation summary.

6 Results on 100 Images

Across the 95 images with maskable cues the adversary attains a country-scale accuracy (error < 750 km) of 77.9%, 45.3% within 25 km, median error 59 km (median $p_{\text{true}} = 0.137$). Attribution is computed on *all* 95 images (the primary result); we additionally stratify into 71 country-correct and 24 country-incorrect images, since a wrong top-1 does not imply zero belief movement (mPL measures odds shift, not top-1 correctness).

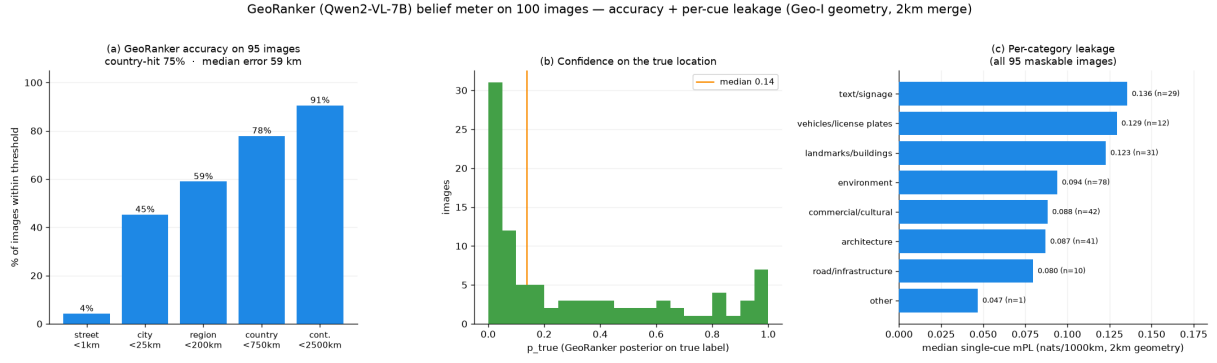


Figure 2: (a) adversary accuracy (error thresholds); (b) confidence on the true label; (c) per-category single-cue mPL (median) — baseline; see Figure 3 for the corrected attribution.

Per-category leakage. As the traditional ablation *baseline*, the raw single-cue masking effect (median mPL, all 95 images) ranks text/signage first (0.136, $n=29$), then vehicles/plates 0.129 (12), landmarks/buildings 0.123 (31), environment 0.094 (78), commercial/cultural 0.088 (42), architecture 0.087 (41), road/infrastructure 0.080 (10). By §3 this over-states per-cue attribution by double-counting shared leakage. Our *primary* result is the Shapley-corrected ranking (Figure 3a): text/signage 0.080 > landmarks/buildings 0.072 > environment 0.060 > vehicles/plates 0.045 > architecture 0.042 > commercial/cultural 0.037 > road/infrastructure 0.034. The correction roughly halves every median and reorders the top: **vehicles/plates falls from 2nd (raw) to 4th** — its apparent standalone importance is mostly shared with co-occurring cues in the same scene, whereas text/signage remains 1st, i.e. genuinely the least-shared leakage source. (Restricting to the 71 country-correct images shifts the top-2 order, confirming that country-conditioning is itself a selection effect; we therefore keep all 95 as the primary set.)

Shapley attribution. The full 2^m lattice is evaluated for every multi-cue image ($m = 2-5$) and augmented with the $m = 1$ images ($\varphi = v(\{1\})$), so all 95 images receive an attribution; the efficiency identity $\sum \phi_k = v(N)$ holds exactly wherever $m \geq 2$. Interactions use the Shapley interaction index of §4 (averaged over coalition contexts), not the empty-context second difference. Across 239 cue pairs (Figure 3b):

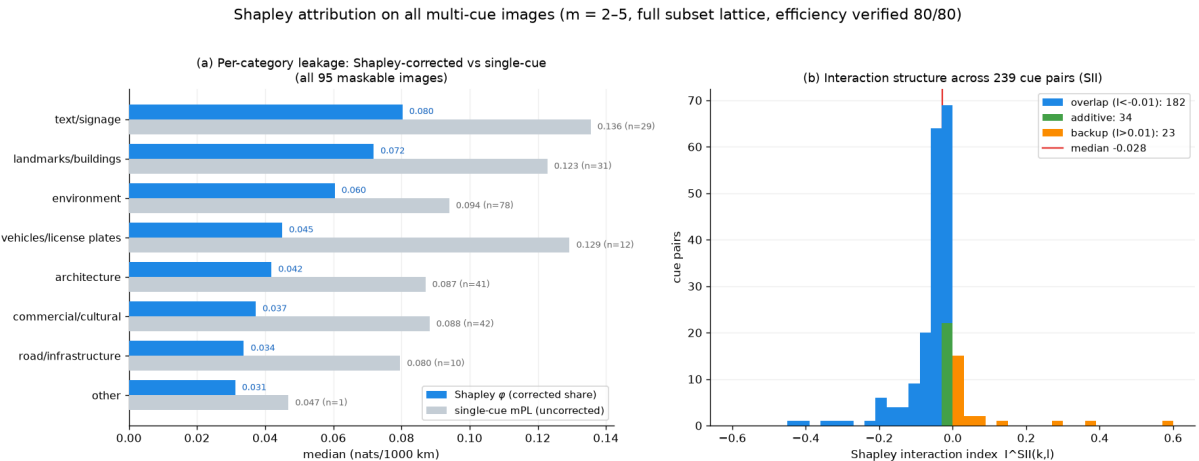


Figure 3: (a) per-category leakage, Shapley-corrected vs single-cue; (b) Shapley-interaction structure across 239 cue pairs.

Interaction type	Condition	Pairs	Meaning
Overlap (sub-additive)	$SII < -0.01$	182	either cue alone does most of the movement
$\approx$ Additive	$ SII \leq 0.01$	34	independent contributions
Backup (super-additive)	$SII > 0.01$	23	cues must be masked jointly

Table 2: Shapley-interaction structure across 239 cue pairs (all maskable images; median SII = -0.028). The empty-context second difference over-states overlap (215/12/12).

Overlap dominance (182/239, 76% by the proper SII) shows that single-cue mPL systematically *double-counts shared leakage and therefore overstates per-cue attribution*; Shapley redistributes the shared mass so that per-cue contributions sum exactly to $v(N)$. (In the 23 backup pairs, $SII > 0$, the single-cue value errs oppositely and understates the contribution — note the proper SII surfaces roughly twice as many backup pairs as the empty-context index, which had masked them under full context.) Extremes: Lake Bled is the strongest backup pair (church on island + mountains); the strongest overlap pairs are architecture-vegetation combinations. Only 2 of 231 cues have $\varphi < -0.01$.

References

- [1] L. S. Shapley. A Value for n-Person Games. 1953.
- [2] P. Jia, S. Park, S. Gao, X. Zhao, Y. Li. GeoRanker: Distance-Aware Ranking for Worldwide Image Geolocalization. NeurIPS 2025.
- [3] Meta AI. SAM 3: Segment Anything with Concepts. 2025.
- [4] Chen et al. Metric-normalized posterior leakage (mPL). 2026. (Full citation as in the companion proposal.)